1. Analysis of Published Accounts

1.1. Introduction

- Published accounts reveal some details into a company's performance, such as yearly changes in profits, revenue, current assets, liabilities, and shareholders' equity. Comparisons can also be made between companies.
- However, individual accounting figures alone provide limited insight into performance or efficiency.
- Accountants use ratios - comparing two figures - to create meaningful assessments.
- These ratios fall into five main categories, starting with liquidity ratios.

1.2. Liquidity ratios

The meaning and importance of liquidity

- **Liquidity** measures how easily a business can meet its short-term debts or liabilities.
- **Liquidity ratios** are an important measure of a business's ability to continue trading.
  - **Working capital** is essential for liquidity. Insufficient working capital can lead to illiquidity, inability to pay debts, loss of supplier trust, and denied bank loans
  - Excess **working capital** ties up funds that could be invested more profitably elsewhere.
- **There are two liquidity ratios:**
  - Current ratio
  - Acid-test ratio

Current ratio: calculation and interpretation

- The formula for current ratio is:
  - **Current ratio =** $\frac{\text{current assets}}{\text{current liabilities}}$
- The result is generally expressed as a ratio *(For e.g 2:1)*
  - This result indicates that the business can cover its short-term debts **twice**, indicating **strong liquidity**
- There's no universal benchmark for liquidity. Accountants often suggest 1.5:1 to 2:1, but it depends on the industry and trends.
  - A 1.5:1 ratio may raise concern if it was significantly higher previously.
- Review the accounting data for two sports clothing manufacturers, Port Louis Sports (PLS) and Bengaluru Sports Kit (BSK), in the table below.

	Current assets	Current liabilities	Current ratio
PLS	100	50	100/50 = 2:1
BSK	720	720	720/720 = 1:1

- **Points to note**
  - **PLS** is in a safe liquidity position, with **2 dollars** in current assets for every **1 dollar** of liabilities, ensuring stability.
  - **BSK's** liquidity is more concerning, as it only has **1 dollar** in assets per **1 dollar** of debt. This could cause issues if creditors demand repayment or if assets are not easily converted to cash.
  - Low current ratios are normal in industries like food retailing, where steady cash inflows cover short-term debts.
  - Ratios over 2:1 might indicate excess funds tied up in inventories, receivables, or cash, which could be better invested in productive assets like equipment.
  - A low ratio may require management to boost current assets.

Acid test ratio

- The acid test ratio AKA quick ratio, is a stricter test of a firm's liquidity and is measured by:
  - **Acid test ratio =** $\frac{\text{liquid assets}}{\text{current liabilities}}$
  - **Liquid assets = current assets - inventories**
- The acid test ratio **excludes inventories**, the least liquid current assets, as they may not sell quickly.
  - This provides a clearer view of the business's ability to pay short-term debts.
- Using the two businesses PLS and BSK again, look at the table below

	Liquid assets	Current liabilities	Acid test ratio
PLS	50	50	1:1
BSK	540	720	0.75:1

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- Points to note
  - Results below 1:1 indicate less than $1 of liquid assets per dollar of short-term debt, suggesting BSK may face liquidity issues. Trends over recent years offer better context.
  - For instance, if last year's acid test was 0.5:1, a current 1:1 ratio shows improved liquidity, whereas a drop from 1:1 signals a decline.
  - Businesses with high inventory levels, like furniture retailers, often have significant differences between current and acid test ratios, which may not be concerning. However, for industries like tech manufacturing, where inventories lose value quickly, this could be problematic.
  - Selling inventory for cash won't change the current ratio but will improve the acid test ratio, as cash is more liquid than inventory.

Methods of improving liquidity

Methods to increase liquidity	Examples	Evaluation of method
Sell fixed assets for cash and lease them back if needed.	Land and property sold to a leasing company	Selling assets quickly may undervalue them, and leasing needed assets increases overheads, reducing the operating profit margin.
Sell inventories for cash	Inventories of finished goods sold off discount raising cash.	Selling inventories at a discount reduces the gross profit margin and may damage the brand image.
Use JIT inventory management.	JIT inventory management will clear inventories held.	Inventories may be needed for changing demand, and JIT could be difficult to implement in some industries.
Increase loans to boost cash and working capital.	Take long-term bank loans if prospects are strong	These will increase the gearing ratio and higher interest costs will reduce annual profit

1.3. Profitability ratios

The meaning and importance of profitability

- Measuring profit against targets is important, as profit is a major objective for many businesses.
- A high profit does not always indicate high profitability if sales and capital are also high. Consider the profit figures from the two sports clothing manufacturers, PLS and BSK.

	2021 operating profit ($000)
PLS	50
BSK	500

- We cannot assess the performance of the two companies from these results. The only conclusion is that BSK made an operating profit ten times greater than PLS.
- Now look at other information about these two businesses

	2021 Revenue ($000)
PLS	250
BSK	3200

- Additional data provides a clearer picture of the companies' performance, especially when comparing revenue and profit.
- Profitability, which measures profit relative to sales or capital invested, is a better indicator of business performance than overall profit.
  - Higher profitability increases the potential for high returns to investors.
- The three key profitability ratios in the A Level Business syllabus are gross profit margin, operating profit margin, and return on capital employed.

Return on capital employed: calculation and interpretation

- The return on capital employed (RoCE) ratio, often called the primary efficiency ratio, measures profitability by comparing operating profit to invested capital.
- It is measured by:
  - Return on capital employed ratio = $\frac{\text{Operating profit}}{\text{Capital employed}}$ x 100
  - Capital employed = non-current liabilities + shareholders' equity
  - Shareholders' equity = issued shares + reserves

	Operating profit $M	Capital employed $M	Return on capital employed
PLS	50	400	50/400 x 100 = 12.5%
BSK	500	5000	500/5000 x 100 = 10%

- The results show PLS's management is more effective at generating profit from invested capital.
- Points to note:
  - A higher RoCE indicates a greater return on invested capital.
  - Comparing RoCE with other companies and past performance reveals profitability trends.
  - Results can be compared to bank interest rates to assess whether capital could earn more with less risk.
  - RoCE should exceed borrowing costs, as lower returns than interest reduce shareholder gains.
  - RoCE improves through efficient, profitable use of assets purchased with the capital employed.

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Gross profit margin: calculation and interpretation

- The gross profit margin ratio assesses how effectively a business converts revenue into gross profit and is widely used to measure company performance
- It is measured by the formula:
  - Gross profit margin ratio = $\frac{\text{Gross profit}}{\text{Revenue}}$ = 100

	Gross profit $000	Revenue $000	Gross profit margin
PLS	125	250	125/250 x 100 = 50%
BSK	800	3200	800/3200 x 100 = 25%

- Points to note:
  - Despite BSK's higher gross profit, it has a lower gross profit margin due to its revenue being 12.8 times greater than PLS, but gross profit only 6.4 times greater.
  - The lower margin may result from BSK's low-price strategy or higher cost of sales per unit.
  - Higher costs could stem from increased material or direct labor costs, suggestingBSK's management may be less effective at controlling costs.
  - The gross profit margin indicates how well managers add value to sales.
  - Comparing ratios across industries is misleading due to differing risks and profit margins.

(Operating) Profit margin: calculation and interpretation

- The operating profit margin ratio uses operating profit, calculated by subtracting overhead expenses from gross profit. It measures how effectively a business converts sales into operational profit.
- It is measured by the formula:
  - Operating profit margin ratio = $\frac{\text{operating profit}}{\text{revenue}}$ x 100

	Operating profit $000	Revenue	Operating profit margin
PLS	50	250	50/250 x 100 = 20%
BSK	500	3200	500/3200 x 100 = 15.6%

- Points to note:
  - The profitability gap between BSK and PLS has narrowed, with a smaller difference in operating profit margins despite a substantial gap in gross profit margins.
  - BSK's higher overhead expenses result in lower operating profit relative to revenue.
  - BSK could improve by reducing overheads or increasing revenue without raising expenses.
  - Comparing results over time would reveal trends in performance and profitability, offering a clearer view of management effectiveness.
  - The operating profit margin and its trend over time reflect how well management converts revenue into profit after expenses.

Methods of improving profitability

Methods	Examples	Evaluation of method
Reduce direct costs	Use cheaper materials. Cut labour costs (Example: offshoring production to low cost countries). Cut labour costs through automation. Cut wage costs by reducing them	Quality and therefore, brand image may be damaged. Communication issues may occur. Machinery will increase overhead costs, workers need re-training, and short-term profits will decrease. Motivation may fall.
Increase prices	Increase profit per item sold	Total profit may drop if many consumers switch to competitors. Links to PED
Increase profit margin (reducing overheads/reducing interest costs)	Relocate to low-cost site. Cut promotion costs. Delayer the business. Cut long-term borrowing may be by raising finance from new issue of shares.	Lower rent may mean moving to a less attractive area, harming the image. Cutting promotion costs may reduce sales more than fixed costs. Having fewer managers may reduce efficiency. New share issue's success depends on business prospects and investor confidence.

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Methods to increase RoCE	Evaluation of method
Raise operating profit without raising capital employed. This may be done by **raising prices, reducing direct costs per unit. Reducing overheads, such as delayering or cutting promotion costs.**	However, demand may be price elastic. Cheaper materials means lower quality. Short-run profit may suffer and may have drawbacks such as lower sales due to less promotion.
Reduce capital employed. This may be done by **selling assets contributing nothing or little to sales/profit, and use the capital to reduce debts.**	However, these assets may be required in the future, for example for expansion of the business.

1.4. Financial efficiency ratios

The meaning and importance of financial efficiency

- Various ratios assess how efficiently management uses business resources. High financial efficiency means effective asset use, minimizing borrowing needs and financing costs.
- The key financial efficiency ratios are **inventory, receivables, and payables turnover.**

Rate of inventory turnover: calculation and interpretation

- Lower capital tied up in inventories is best for the business.
- Inventory turnover measures how often inventory is **bought and resold in a period**.
- An increasing ratio indicates reduced inventory investment and better financial efficiency. If inventory is bought once a year, turnover would be 2, with high inventory investment.
  - **Rate of inventory turnover** = $\frac{\text{Cost of sales}}{\text{Average inventory}}$
  - **Average inventory** = $\frac{\text{Opening inventory} + \text{Closing inventory}}{2}$

	Cost of sales $M	Average inventory $M	Rate of inventory turnover
PLS	125	40	125/40 = 3.125
BSK	2400	200	2400/200 = 12

- The table above shows average inventory figures for the year, not the difference between current and liquid assets.
- BSK's inventory turnover rate reflects more efficient inventory management, with lower inventories relative to cost of sales than PLS.
- Adopting a JIT system could help PLS reduce inventory levels and improve turnover rates.
- **Points to note:**
  - Inventory turnover is measured in the number of times inventory is sold in a year, not as a percentage.
  - A higher turnover indicates efficient inventory management, with JIT systems achieving high rates.
  - Turnover varies by industry; for example, fresh fish retailers typically have higher rates than car dealers, making cross-industry comparisons unreliable.
  - This metric is irrelevant for service firms like insurance companies that do not hold inventory.

Trade receivables turnover (days): calculation and interpretation

- **Trade receivables turnover (days)** measures the average time to collect payments from credit customers. Shorter periods indicate better working capital management.
  - **Trade receivables turnover** = $\frac{\text{Trade receivables}}{\text{Credit sales}}$ x 365

	Trade receivables	Revenue	Trade receivables turnover
PLS	45	250	45/250 x 365 = 65.7 days
BSK	500	3200	500/3200 x 365 = 57 days

- Both companies allow about two months for customers to pay, likely due to competition in the sports clothing market. Longer credit periods may attract orders but increase working capital needs.
- **Points to note:**
  - Trade receivables turnover varies by industry, with low turnover often reflecting stricter credit policies.
  - High turnover may attract customers with extended credit but could also indicate poor management of receivables and repayment periods.
  - Shortening credit terms, such as reducing from 60 to 30 days, or refusing credit to late payers, can improve results. However, such changes must balance sales growth goals with financial control priorities.

Trade payables turnover (days): calculation and interpretation

- **Trade payables turnover (days)** measures the average time a business takes to pay its suppliers. A longer period reduces working capital needs. The formula can use either credit purchases or cost of goods sold.
  - **Trade payables turnover** = $\frac{\text{Trade Payables}}{\text{Credit purchases}}$ x 365

	Trade payables ($)	Credit purchases ($)	Trade payables turnover
PLS	16	100	16/100 x 365 = 58.4 days
BSK	280	1920	280/1920 x 365 = 53.2 days

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- Points to note:
  - BSK pays suppliers faster than PLS, creating a cash flow problem as it needs more working capital.
  - While PLS takes longer to pay suppliers, it still faces the same issue due to delayed customer payments.
  - Both companies could improve financial efficiency.

Methods of improving financial efficiency

- Financial efficiency can improve by reducing the need for high working capital.

Method	Evaluation
Adopt JIT to boost inventory turnover.	Risk of zero-inventory from supply delays.
Reduce customer credit to lower debt collection period.	Customers may choose businesses with longer credit periods.
Delay supplier payments to extend creditor payment period.	Supplier discounts for quick payment may decrease, or they may demand faster payments.

1.5. Gearing ratio

- When a business takes out a loan, it funds expansion but increases business risk.

The meaning and importance of gearing

- The gearing ratio shows how much business capital comes from debt.
- Higher reliance on loans means higher gearing, increasing the risk of repayment issues if profits drop.
- Gearing is crucial to assess.

Gearing ratio: Calculation and interpretation

- The gearing ratio is measured by the formula:
  - Gearing ratio = $\frac{\text{Non-current liabilites}}{\text{Capital employed}}$ x 100

	Non-current liabilities	Capital employed	Gearing ratio
PLS	40	400	40/400 x 100 = 10%
BSK	2000	5000	2000/5000 x 100 = 40%

- PLS relies less on bank loans than BSK, reducing risk but possibly missing profitable opportunities. If profits drop during a recession, BSK may regret its heavy reliance on long-term loans.

- Points to note:
  - The ratio shows how much capital is financed by long-term debt. A result over 50% indicates high gearing.
  - Higher gearing increases shareholder risk due to:
    - Higher debt leads to more interest payments, affecting dividends and retained earnings, especially with high interest rates and low profits.
    - Interest must still be paid even if profits fall.
    - High debt repayment can lead to low liquidity.
  - A low gearing ratio suggests a safer strategy but may disappoint shareholders seeking rapid returns.
  - High-gearing businesses with growth strategies offer faster returns compared to slower-growth, low-gearing companies.

Methods of improving gearing

Methods	Evaluation
Sell shares to raise capital and repay loans.	This dilutes existing shareholders' control. Dividend payments must increase to maintain yield. Poor economic conditions may lead to shares being sold at a low price.
Reduce dividends, retain profits, and use them to repay loans.	Low profits may reduce dividends, leading to lower returns for shareholders
Sell assets to raise finance and repay loans.	Quick asset sales may lower prices, reducing business value and limiting expansion, unless the assets are unnecessary (e.g., an empty office).

1.6. Investment ratios

- Shareholders and potential investors will closely analyze company accounts to assess investment potential.

The meaning and importance of return to investors

- Investment ratios are important to shareholders and potential investors.
- They show potential returns from capital gains (rising share price) and dividends, unless profits are low or losses occur.

Dividend yield: calculation and interpretation

- The dividend yield ratio measures the return a shareholder gets from the dividend per share at the current share price, calculated using this formula:
  - Dividend yield ratio = $\frac{\text{Dividend per share}}{\text{Market share price}}$
  - Dividend per share = $\frac{\text{Total annual dividends}}{\text{Total number of issued shares}}$

	Dividends ($M)	Number of issued shares ($M)	Dividend per share ($)	Market share price ($)	Dividend yield %
PLS	21	140	0.15	$1.50	$0.15x100/$ 1.5 = 10%
BSK	140	200	0.70	$10.00	$0.7x100/10.$ = 7%

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- Shareholders in PLS earn a higher return than those in BSK.
- Potential investors should consider other ratios alongside the dividend yield before making a decision.

Points to note:

- If the share price rises with an unchanged dividend, the dividend yield falls.
- If the dividend increases without a share price change, the yield rises.
- Shareholders compare this return with other investments and similar companies.
- The result should be compared to previous years and industry peers for effective analysis.
- High dividend yields may attract potential shareholders if the share price is stable.
- Directors may pay dividends from reserves to maintain shareholder loyalty, even in low-profit or loss years.
- Directors may reduce dividends to increase retained earnings, supporting business expansion and lowering the gearing ratio.
- A high dividend yield may signal a poor investment if the share price has fallen, indicating market concerns about the company's future.

Dividend cover: calculation and interpretation

- The **dividend cover ratio** shows how many times a company's profit can cover its ordinary dividends. A higher ratio indicates stronger dividend payment ability and allows more profit to be reinvested into the business.
  - **dividend cover ratio** = $\frac{\text{Profit for the year}}{\text{Annual dividends}}$

	Dividends ($)	Profit for the year ($)	Dividend cover
PLS	21	30	30/21 = 1.4
BSK	140	400	400/140 = 2.9

- The dividend cover ratio suggests BSK's directors could have paid a higher dividend but may be retaining profits to fund business expansion, benefiting shareholders in the future.
- **Points to note:**
  - If directors increase dividends without higher profits, the dividend cover ratio would drop, raising concerns about sustainability.
  - A low ratio also suggests minimal profit retention, potentially affecting future expansion.

Price/earnings ratio: calculation and interpretation

- The **price/earnings (P/E)** ratio is essential for shareholders and reflects investor confidence in a company's future prospects. A high P/E ratio indicates expectations of higher earnings growth, while a ratio of 1 suggests **low confidence** in future earnings.
  - **P/E ratio** = $\frac{\text{Market share price}}{\text{Earnings per share}}$
  - **Earnings per share** = $\frac{\text{Profit for the year}}{\text{Number of shares issued}}$

	Current share price ($)	Earnings per share ($)	Price/earnings ratio
PLS	1.50	0.21	7.1
BSK	10.00	2.00	5.0

- **Points to note:**
  - The P/E ratio reflects investor confidence, with PLS's ratio of 7.1 indicating investors pay 7.10 dollars for each dollar of earnings and expect a 7.1-year payback at current earnings levels.
  - The ratio should only be compared within the same industry, as optimism varies across sectors.
  - Comparing the P/E of a high-growth industry, like technology, to a low-growth one, like utilities, is not meaningful due to differing growth prospects.

Methods of improving investor return

- Increasing profits improves returns to investors by enabling higher dividends and raising dividend yield.
- Business growth and profitability also push up share prices, benefiting shareholders through both dividends and capital gains.
- **However, there is a short-term trade off:**
  - Increasing profit may require investing in assets.
  - Reducing dividends improves retained profit and internal finance.
  - This lowers short-term shareholder returns.
  - Successful expansion raises profits, benefiting dividends and share prices, but only in the long-term.

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